Name: Khrishi Parmar -21027654
Module code: UFMFFK-15-2 Flight

Regional Air freight aircraft POH



Specification

Measurements	Value
Fuselage Length	24.2m
Wingspan	19.25m
Wing Area	32.51m2
Cabin length	14.1m
Performance	
OEW	8812kg
MTOW	16020kg
Max Payload	5290kg
Max Fuel	3600kg
Design Fuel	1918kg
Fuel	Assumed equally distributed in the wing area
MLW	15570kg

Engines	Value
Weight	2x450kg
Engine power	1625kW per engine ISA at sea level
Engine Specific fuel consumption	0.485 lb/hp/hr
Static Thrust per engine	18550N ISA
Propeller Diameter	3.5m, variable pitch
Propeller efficiency	Function of Mach Number

Aircraft masses	Value
Wings	1573.6kg
Horizontal Tail	334.5kg
Vertical fin	289.9kg
Fuselage	5714kg

Landing Gear Loading limits	Value
Front	370kg
Main	18000kg

Parameter	Value
Datum	Tip of root chord at centreline
Crew	2
OEW c.g	1.525m aft of datum

Vertical Tail

Parameter	Value
chord profile	NACA-0015
Rudder	35% local Chord
Taper Ratio	0.6
Max Deflection	+-20°

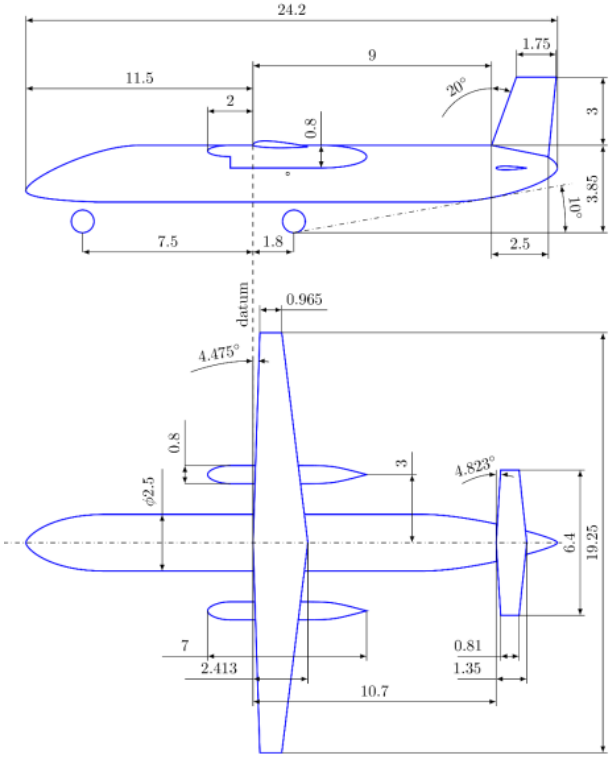
Wings

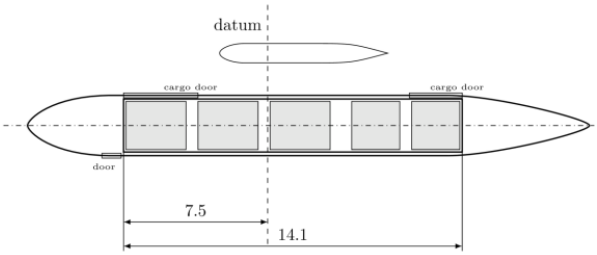
Parameter	Value
Wing chord profile	64-210
Rigging Angle	2° relative to fuselage centreline
Washout (root to tip)	2°
High Lift devices	Plain Flaps 35° Max deflection 30% of local Chord. Take-off setting 20°
Dihedral	0°

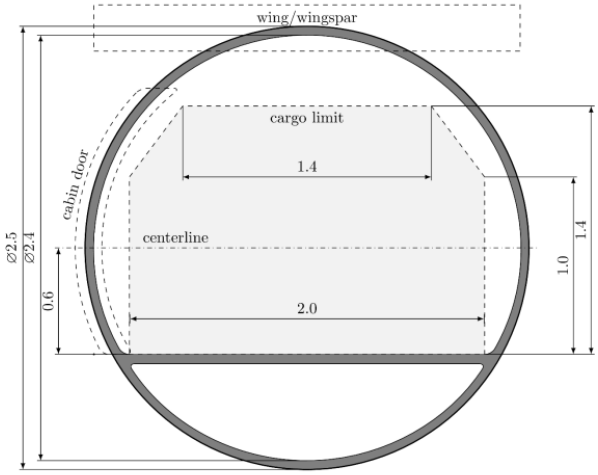
Horizontal Tail

Parameter	Value
chord profile	NACA-0012
Rigging angle	0° relative to fuselage centreline
Elevator	50% of Tail plane area
Dihedral	0°
Gearing Ratio	3.5rad/m
Max Deflection	+-20°

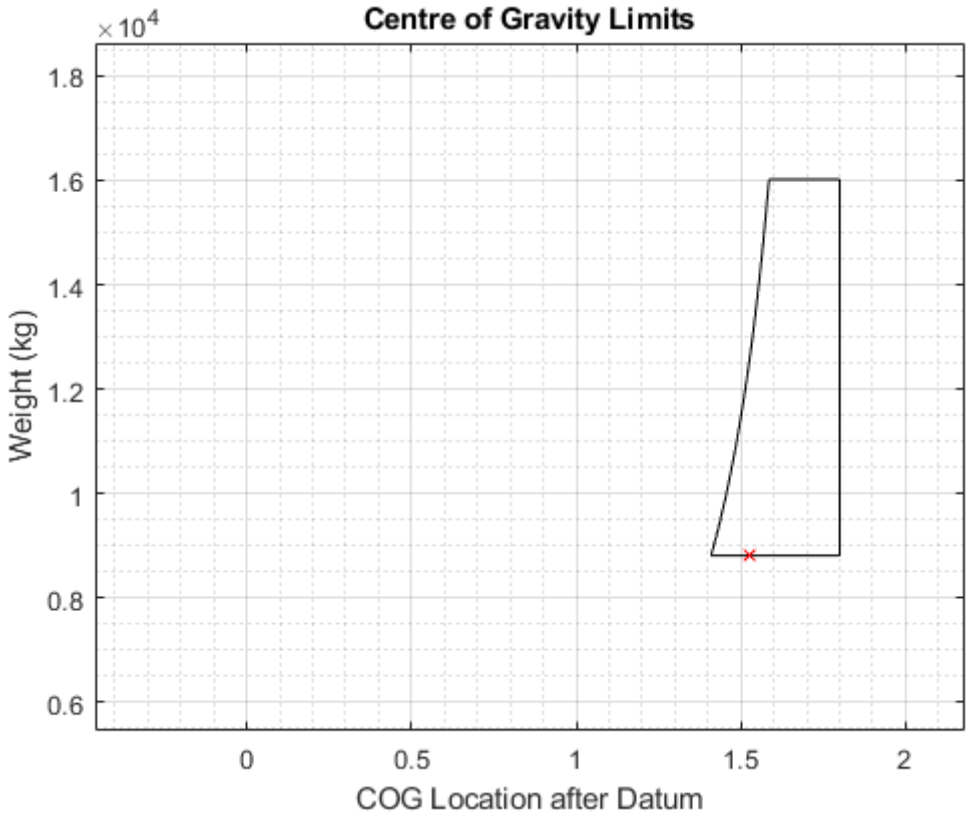
Drawings







Centre of Gravity Limits

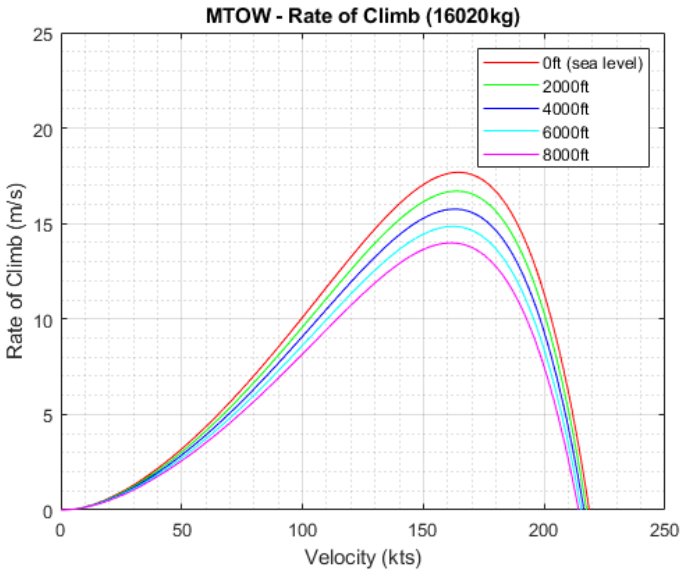


Take-off Distance Table

Take off MTOW (16020kg)			
Altitude (ft)	Ground Run (ft)	Lift air Phase (ft)	Total Take off Distance (ft)
0 (Sea Level)	2.6147e+03	1.866e+03	9.037e+03
2000	7.6060e+03	1.9792e+03	9.585e+03
4000	8.0743e+03	2.1010e+03	1.018e+04
6000	8.5785e+03	2.2322e+03	1.081e+04
8000	9.1220e+03	2.3736e+03	1.1496e+04

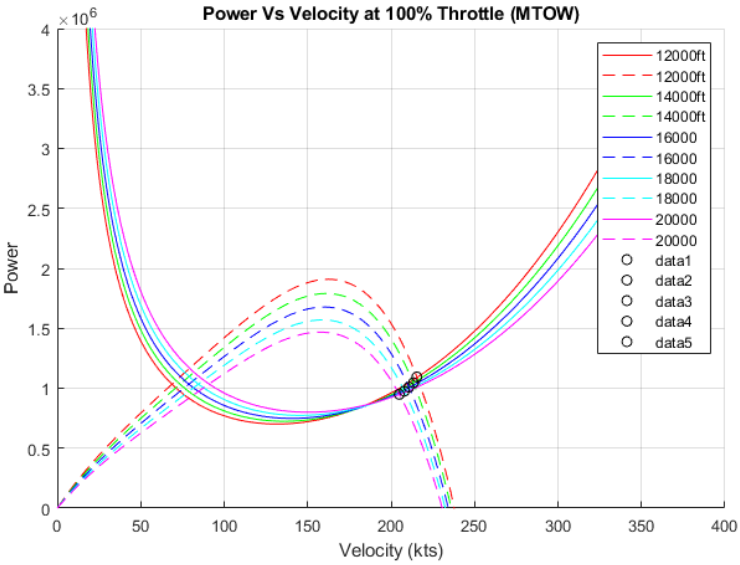
Climb Performance Table

Rate of Climb – MTOW (16020kg)	
Altitude (ft)	**Rate of Climb (m/s)**
0 (Sea Level)	17.6
2000	16.6
4000	15.7
6000	14.85
8000	13.9

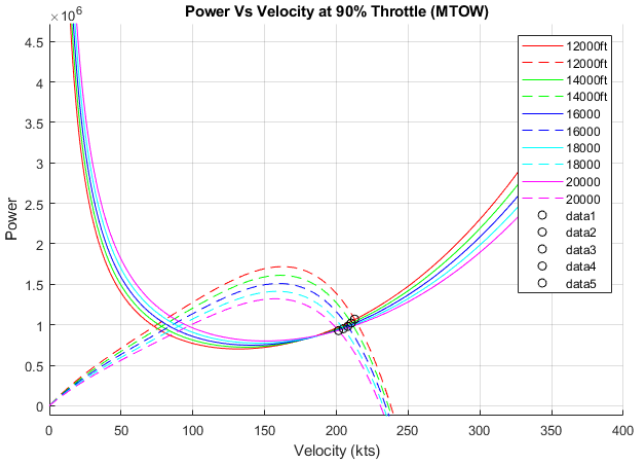


Cruise Performance Table

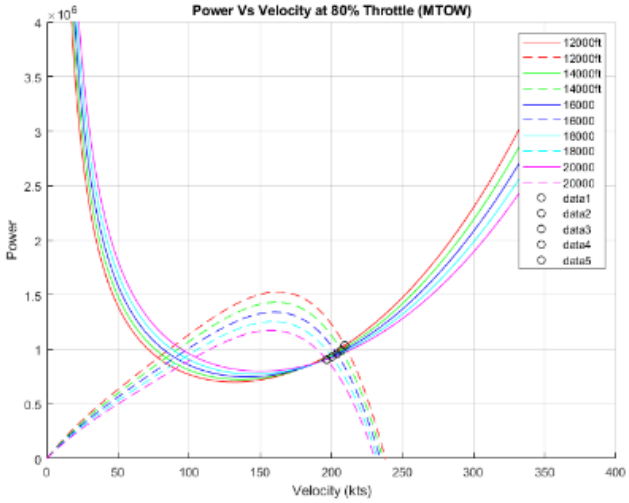
Cruise at 100% Throttle		
Altitudes (ft)	**Cruise velocity (kts)**	**Fuel consumption (kg/hr/W)**
12000	215.435	1093750
14000	213.393	1045790
16000	210.671	1008820
18000	207.948	975501
20000	204.885	949275



Cruise at 90% Throttle		
Altitudes (ft)	**Cruise velocity (kts)**	**Fuel consumption (kg/hr/W)**
12000	209.309	1037320
14000	206.927	991063
16000	203.864	959046
18000	200.46	931142
20000	196.717	905713



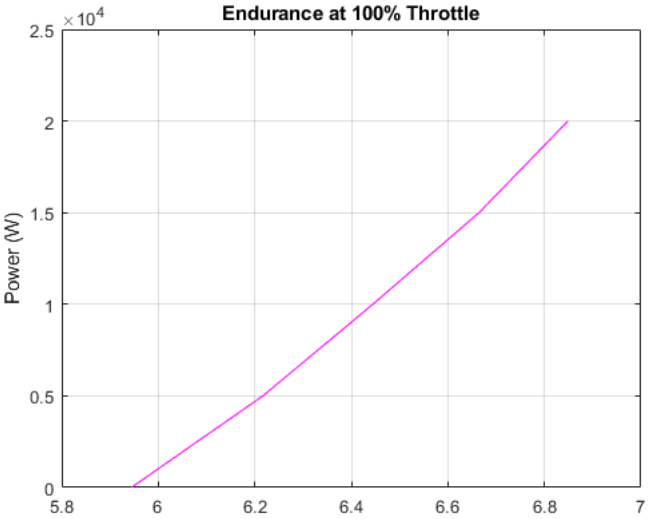
Cruise at 80% Throttle		
Altitudes (ft)	**Cruise velocity (kts)**	**Fuel consumption (kg/hr/W)**
12000	212.713	1069570
14000	210.33	1021140
16000	207.948	980609
18000	204.885	952172
20000	201.481	928281



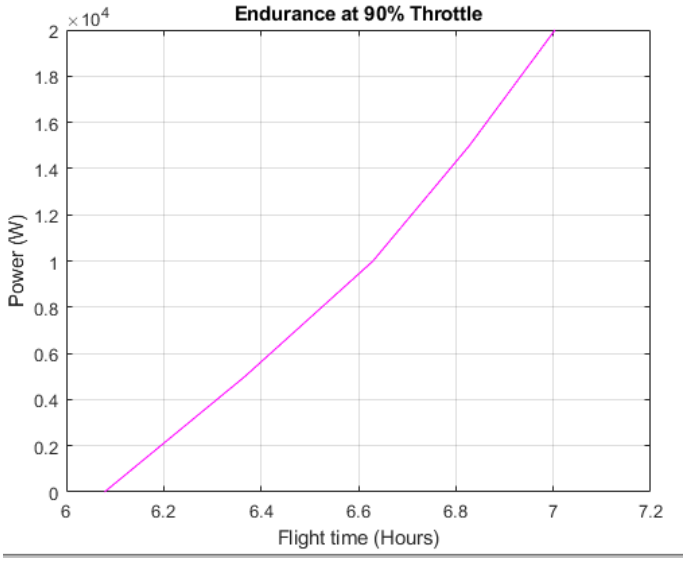
Landing Performance Table

Take off MLW (15570kg)			
Altitude (ft)	Ground Deceleration (ft)	Land air Phase (ft)	Total Landing Distance (ft)
0 (Sea Level)	2.4374e+03	2.8333e+03	5.2708e+03
2000	2.5854e+03	2.9308e+03	5.5162e+03
4000	2.7445e+03	3.0356e+03	5.7802e+03
6000	2.9159e+03	3.1484e+03	6.0644e+03
8000	3.1007e+03	3.2703e+03	6.3710e+03

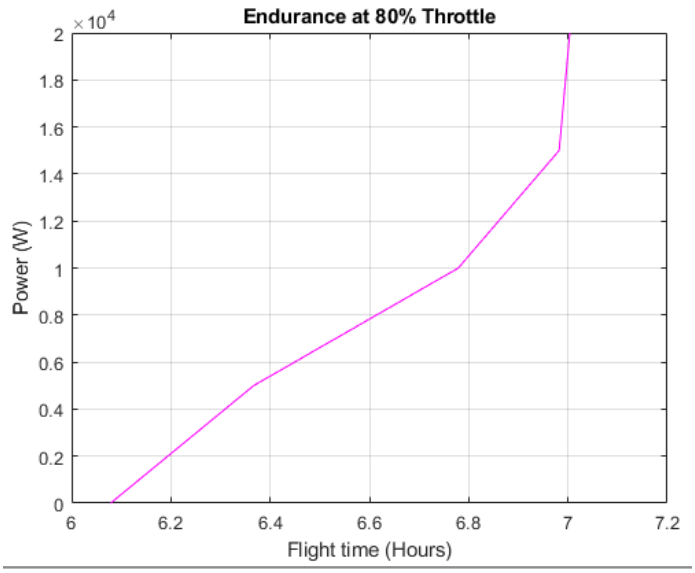
Endurance Profile



Endurance at 100% Throttle (MTOW)	
Altitude (ft)	**Endurance (Hours)**
12000	5.94411394766300
14000	6.21671141458266
16000	6.44453384177198
18000	6.66465193808761
20000	6.84877894209413



Endurance at 90% Throttle (MTOW)	
Altitude (ft)	**Endurance (Hours)**
12000	6.07849381551128
14000	6.36678088240242
16000	6.62993571368038
18000	6.82794141211504
20000	7.00367090380650



Endurance at 80% Throttle (MTOW)	
Altitude (ft)	**Endurance (Hours)**
12000	6.07849381551128
14000	6.36678088240242
16000	6.77900187296168
18000	6.98215162698751
20000	7.00367090380650